

Trends in atomic and ionic radius

Learning outcomes

By the end of this section you should be able to:

- Outline what defines the atomic radius.
- Explain the periodicity of atomic and ionic radii.
- Compare the atomic and ionic radii for isoelectronic species.

Defining atomic radius

Models of the atom have evolved throughout time with the most recently updated model being the quantum model. Some of these models are shown in **Figure 1**.

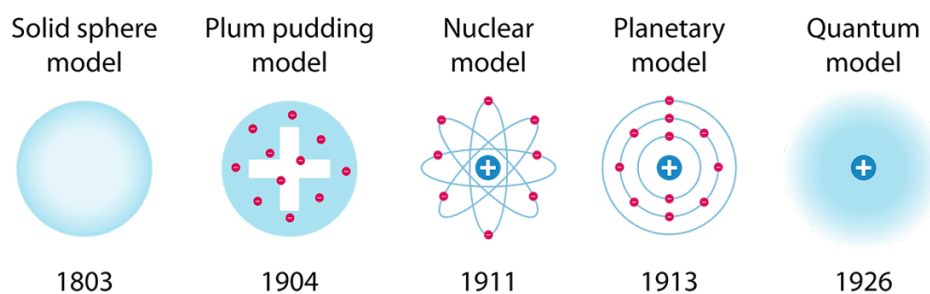


Figure 1. Models of the atom over time.

 [More information for figure 1](#)

The image shows a sequence of atomic models illustrating the evolution of the atom's understanding over time. From left to right, each model depicts a different structure:

1. **Thomson's Plum Pudding Model:** This model shows a light blue circular shape with small red dots scattered within it, representing electrons within a positively charged 'soup.'
2. **Rutherford's Model:** Consists of a dense central positive charge depicted as a cross or nucleus with small red dots (electrons) scattered around it.
3. **Bohr's Model:** Illustrates a central nucleus represented by a blue circle with electrons following distinct circular orbits around it, shown with pink curved arrows.
4. **Electron Cloud Model:** Displays a central nucleus with a surrounding blue faded circle indicating zones of high electron probability, without precise paths.

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In section S1.3.3–4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>) you learned how electrons in atoms occupy atomic orbitals within sublevels, this would be considered the quantum model. In this section we will be looking closely into the sizes of atoms and ions. Looking at the quantum model shown, consider how you would define and describe the size of the atom. What are some ways we describe the size of spherical objects? Do you have any concerns about defining the size of an atom based on the quantum model? How would you expect the sizes of atoms to change as the number of electrons in the outermost energy level change? How would you expect the sizes of atoms to change as the number of occupied energy levels change? Radius is a common parameter used to quantify the size of circles and spheres. For both a circle and a sphere the radius is the distance between the centre to its outer edge. For an atom, what would be at its centre? What would be its outer edge? You may have noticed in the quantum model that the outer edge of an atom is poorly defined. This means that the radius of an atom cannot simply be defined as the distance between the centre to its outer edge, we must define it in another way.

Looking at the quantum model, what part of the atom has a well-defined location? The nucleus! This means it would be easy to identify the nucleus of adjacent atoms in order to deduce the atomic radius of each atom. **Figure 2** illustrates how atomic radius can be defined as half of the distance between two adjacent nuclei.

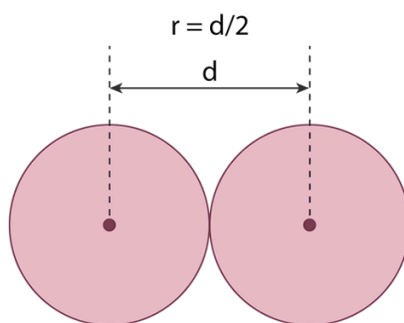


Figure 2. Defining atomic radius as half the distance between two adjacent nuclei.

 More information for figure 2

The image is a diagram illustrating the concept of atomic radius. It displays two adjacent circles, each representing an atom. At the center of each circle is a small dot representing the nucleus. Dotted lines extend from each nucleus to a point between the two atoms, labeled as 'distance between nuclei.' The atomic radius is highlighted as half of this distance, showing how it is measured from the nucleus to the outer edge of the atom. The diagram visually explains that the atomic radius is defined as half the distance between the nuclei of two adjacent atoms.

[Generated by AI]

Effective nuclear charge

The radius of an atom will depend on the interaction of the protons in the nucleus and the electrons in the outer valence shell. This interaction is due to the effective nuclear charge. Effective nuclear charge is the net positive charge experienced by outer valence shell electrons from the force of attraction between protons in the nucleus and the force of repulsion between inner core electrons. The repulsion of the inner core electrons shield the outer electrons from experiencing the full attractive force to the protons in the nucleus. These attractive and repulsive forces are illustrated in **Figure 3**.

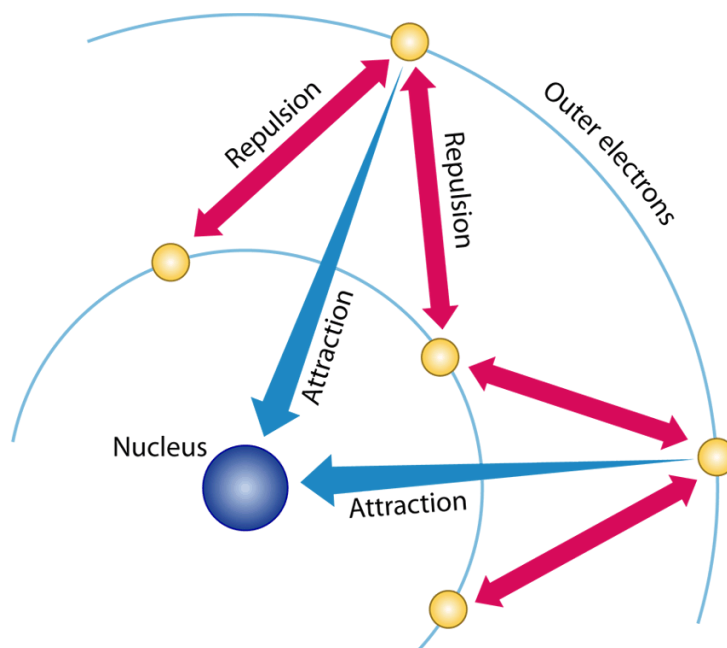


Figure 3. Interactions of the outer shell electrons with the nucleus and inner electrons.

 More information for figure 3

The diagram illustrates the interactions between the outer shell electrons and the nucleus along with the inner electrons in an atom. The central part of the diagram represents the nucleus, where protons are located, shown as a large blue sphere. Surrounding it are gold spheres representing inner core electrons. These are positioned close to the nucleus, indicating the repulsion forces that shield the outer valence electrons from the nucleus' full positive charge.

The outer area depicts pink-colored regions with black arrows indicating the path and forces acted upon the outer shell electrons (not visually represented). The diagram uses upward and diagonal arrows to show the paths of the attractive forces towards the nucleus and repulsive forces from inner electrons. The interactions are designed to display the effective nuclear charge—net positive charge experienced by outer valence electrons due to the balancing act of attraction to the nucleus and repulsion from inner electrons.

[Generated by AI]

As we move across a period, elements have a greater number of protons in the nucleus and a greater number of electrons occupying the outermost energy level. This greater number of protons and valence electrons increases the effective nuclear charge. The greater the effective nuclear charge, the stronger the force of attraction between outer valence electrons and the nucleus. The lower the effective nuclear charge, the weaker the force of attraction between valence electrons and the nucleus.

The worked example looks at the effective nuclear charge of elements in period 3.

Worked example 1

Order the following period 3 elements in order of increasing effective nuclear charge (lowest first).

Mg, Cl, P, S, Si, Na

We can compare the number of protons in the nucleus and the number of valence electrons. This can help us to deduce elements that would have stronger attractions between the nucleus and those with weaker attractions.

	Mg	Cl	P	
<i>protons</i>	12	17	15	
<i>valence electrons</i>	2	7	5	

The order of increasing effective nuclear charge for elements in the same period would correlate to the order of increasing number of protons and increasing number of valence electrons.

$\text{Na} < \text{Mg} < \text{Si} < \text{P} < \text{S} < \text{Cl}$

Trends in atomic radii

The values for the atomic radii of elements in the periodic table can be found in section 10 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-and-ionic-radii-of-the-elements-id-45112/>) of the DP chemistry data booklet. The atomic radius is plotted for the elements in the first four periods in **Figure 4**. Using this graph, identify the elements from the same group (e.g. group 1 is H ($Z = 1$), Li ($Z = 3$), Na ($Z = 11$), K ($Z = 19$)). How would you describe the trend in atomic radius going down a group? Can you think of an explanation for this trend? How would you describe the trend in atomic radius across a period? Can you think of an explanation for this?

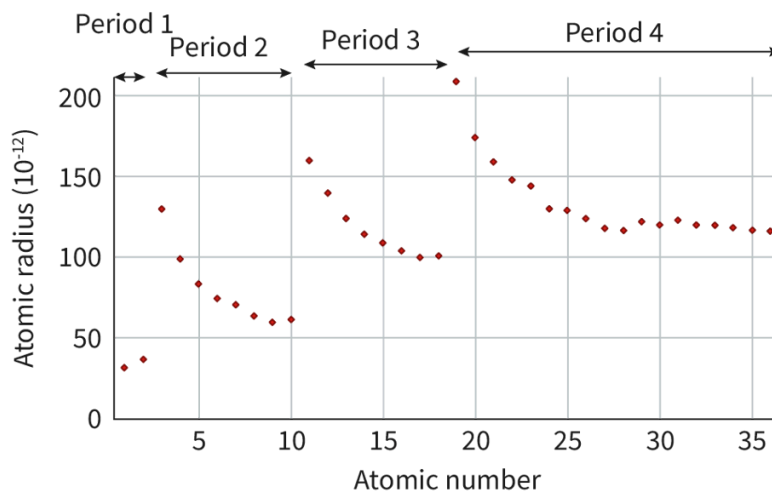


Figure 4. Atomic radii for elements in the first four periods.

 More information for figure 4

The graph displays atomic radii for elements plotted against their atomic numbers for the first four periods of the periodic table. The X-axis represents the atomic number, ranging from 0 to around 35, while the Y-axis represents the atomic radius in picometers (10^{-12}), with a scale from 0 to 200. Data points show a downward trend in atomic radius across a period, with distinct drops as atomic numbers increase, indicating how radius decreases across periods. The trend slightly increases when moving to a new period, reflecting changes in atomic structure. Periods are labeled 1 through 4 above the graph, marking transitions between them.

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Study skills

It can be helpful to create plots using atomic numbers on the x-axis and property of interest on the y-axis to study periodicity. Scatter plots and column graphs are typically used. These types of graphs are very useful to visually represent trends across a period and identify parts of the graph that correlate to the same group.

General trend in atomic radius down a group

Table 1 shows the atomic radius for elements in groups 1, 2, 17 and 18. You will notice that going down a group in the periodic table the atomic radius increases. This is because elements further down in the periodic table have electrons that occupy higher energy levels. These higher energy levels are found further from the nucleus, resulting in elements located further down in a group having larger atomic radii than those higher up in a group. This trend is illustrated using the Bohr model of the atom to visualise energy levels in **Figure 5**.

Table 1. Atomic radius for elements in groups 1, 2, 17 a

Period	Highest occupied energy level	Group 1 Atomic radius ($\times 10^{-12}$ m)		Group 2 Atomic radius ($\times 10^{-12}$ m)		Group 17 Atomic radius ($\times 10^{-12}$ m)
1	$n = 1$	H	32			
2	$n = 2$	Li	130	Be	99	F
3	$n = 3$	Na	160	Mg	140	Cl
4	$n = 4$	K	200	Ca	174	Br
5	$n = 5$	Rb	215	Sr	190	I
6	$n = 6$	Cs	238	Ba	206	At

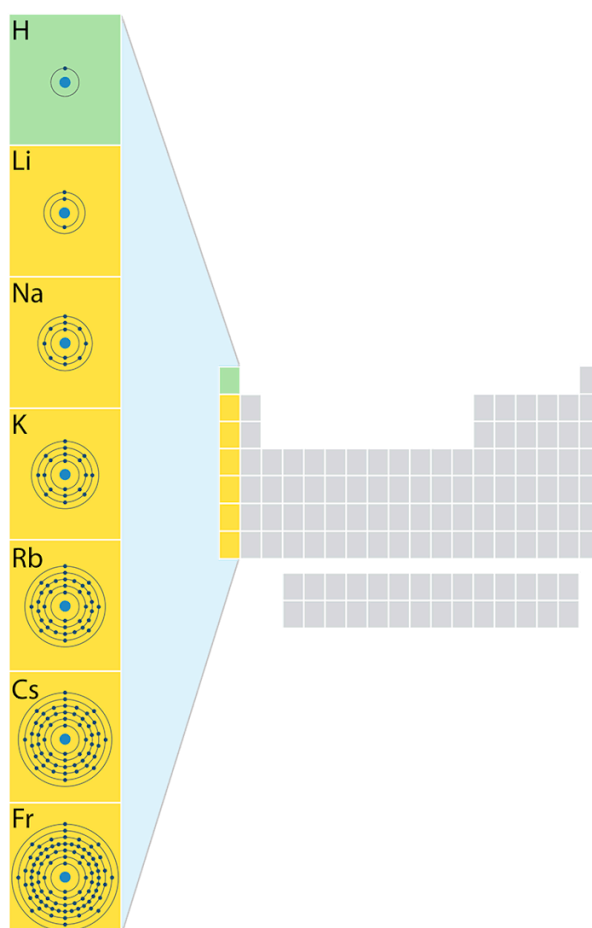


Figure 5. The atomic radius of group 1 atoms looking at the energy levels.

 More information for figure 5

The image is a diagram showing the atomic models of Group 1 elements alongside a portion of the periodic table. The elements hydrogen (H), lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and francium (Fr) are depicted with their

atomic models arranged vertically.

Each element's atomic model shows the nucleus and electron rings. Hydrogen has one electron in a single shell. Lithium has two electron shells with the first ring containing two electrons and the second shell one electron. Sodium's model has three shells, with two electrons in the first, eight in the second, and one in the outer shell. Potassium's electron configuration follows a 2-8-8-1 pattern across its rings. Rubidium, cesium, and francium follow a similar increasing pattern, each having an additional shell and increasing electron count in the inner shells.

Adjacent to these atomic models is a simplified representation of the periodic table. The section showing Group 1 is highlighted, with the elements positioned in a column. Gray squares represent other parts of the periodic table, emphasizing the place of Group 1 elements especially within periods and groups.

[Generated by AI]

General trend in atomic radius across a period

Table 2. Atomic radius for period 3 elements

Element	Na	Mg	Al	Si	P	S
Z_{eff}	1	2	3	4	5	6
Atomic radius ($\times 10^{-12}$ m)	160	140	124	114	109	104

Table 2 shows the atomic radius for period 3 elements. Looking at this table, **Figure 4** and **Table 1** you should notice the general trend of decreasing atomic radius for elements going across a period. This is in contrast to the trend going down a group where a greater number of inner core electrons resulted in a higher atomic radius. As the number of electrons in the outermost energy level increases, the radius decreases.

Why doesn't the atomic radius always increase with an increasing number of electrons? Why would the atomic radius decrease across period 3 when electrons are all occupying the same $n = 3$ energy level? This is due to the increasing effective nuclear charge increasing the attraction of the electrons in the $n = 3$ energy level to the nucleus resulting in a slight decrease of radius of the atom. This trend is illustrated in **Figure 6**.

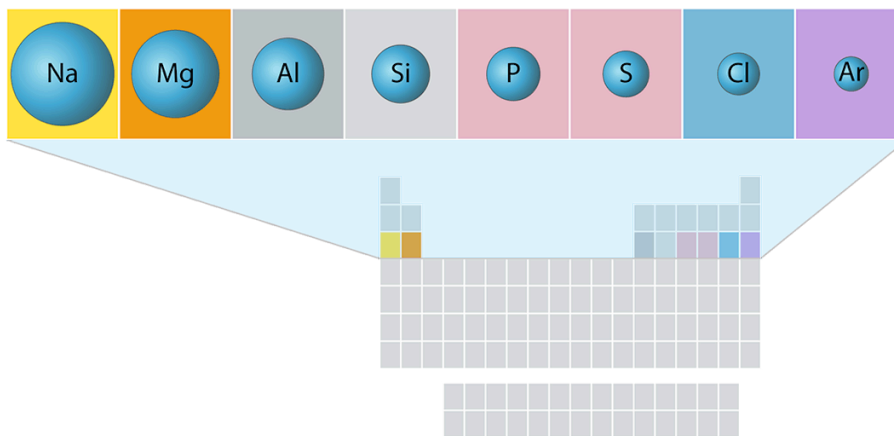


Figure 6. Atomic radius of period 3 atoms.

 More information for figure 6

The image is a diagram illustrating the atomic radius of period 3 elements in the periodic table. It highlights each element with colored blocks and circles representing their relative sizes. The elements included are Sodium (Na), Magnesium (Mg), Aluminum (Al), Silicon (Si), Phosphorus (P), Sulfur (S), Chlorine (Cl), and Argon (Ar). Each element is shown as a circle positioned within its respective block, with the size of the circle indicating the atomic radius. This visual representation helps to convey how the atomic radius decreases across period 3 as effective nuclear charge causes a greater attraction of electrons toward the nucleus. The elements are presented in order from left to right, correlating with their position in the periodic table, demonstrating the trend of decreasing atomic radius across the period.

[Generated by AI]

Study skills

It is important to note that the magnitude of trend is different going down a group than going across a period. The trend in atomic radius for elements going down a group is a **large increase** in size due to the addition of an energy level. The trend in atomic radius going across a period is a **slight decrease** in size due to the increased attraction of the nucleus to the outer shell from the same energy level.

There is periodicity of atomic radii as these general trends repeat throughout the periodic table. **Figure 7** provides an illustration to help visualise the trends in atomic radii throughout the entire periodic table.

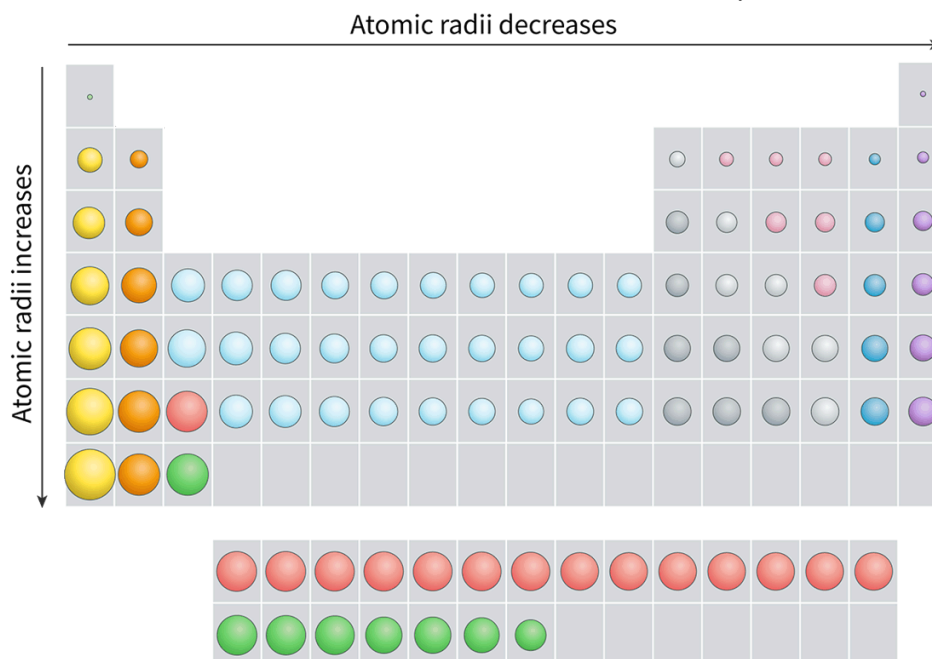


Figure 7. The atomic radii of elements in the periodic table.

 More information for figure 7

The image is a diagram of the periodic table showing the atomic radii of elements. Each element is represented by a colored circle, with the size of the circle indicating the relative atomic radius. The elements are arranged in periods and groups as in the standard periodic table layout. The top left corner has large yellow circles indicating larger atomic radii, which get progressively smaller as you move to the right across a period, then increase again when starting a new period. This pattern repeats across the table, demonstrating the periodic trend of decreasing atomic radii across a period and increasing down a group.

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Aspect: Patterns and trends

In this section we are discussing general trends. You may have noticed that some adjacent elements do not fit these trends. For example, argon has a slightly larger atomic radius than chlorine. The variations from the general trend for atomic radius is largely due to the way that atomic radius is defined in terms of half the internuclear distance between two adjacent nuclei. Since noble gases have a high tendency to exist as single atoms (monoatomic) the radius is based off of an intermolecular interaction, rather than a chemical bond. This overestimates the radius of the atom since atoms are further apart than if they were covalently or metallically bonded.

In **Interactive 1** there are 5 small clusters of elements. Each cluster contains 3 elements so we call these clusters triads. Try ordering the elements within each triad shown in the periodic table below in order of increasing atomic radii. Use (1) for the smallest radii and (3) for the largest.

Smallest radius in triad Largest radius in triad

☒ Check

Interactive 1. Drag and Drop in Order of Increasing Atomic Radii.

More information for interactive 1

This interactive image presents a section of the periodic table and highlights five small clusters of elements called 'triads'. Each triad consists of three elements and highlighted in different colors, and the users are tasked with arranging them in order of increasing atomic radius.

A set of number buttons 1, 2, 3 are given below the table, allowing users to rank the elements in each triad from smallest to largest atomic radius, by dragging and dropping the numbers. The smallest radius should be labeled as 1, and the largest radius as 3.

Triad 1 has H, Li, and Na. Triad 2 has Ca, Rb, and Sr. Triad 3 has B, Al, and Si. Triad 4 has N, O, and F. Triad 5 has Se, Br, and I.

Atomic radius is the size of an atom. According to the size of the atom, the elements are arranged in the periodic table. Down the column, atoms get bigger; across the row, atoms get smaller.

Read below for the solution:

Triad 1: H- 1, Li- 2, Na- 3.

Triad 2: Ca- 1, Sr- 2, Rb- 3.

Triad 3: B- 1, Si- 2, Al- 3.

Triad 4: F- 1, O- 2, N- 3.

Triad 5: Br- 1, Se- 2, I- 3.

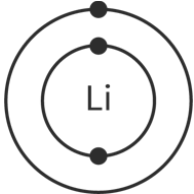
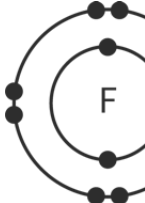
Using these patterns, users can compare atom sizes and arrange the elements in each triad from smallest to largest.

Ionic radii

Recall from section S2.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) how positively charged cations are formed through the loss of electrons and negatively charged anions are formed through the gain of electrons. **Table 3** shows the electron configuration and Bohr model for atoms and ions of lithium and fluorine. By comparing the electron configuration of Li and Li^+ how do you think the atomic radius of Li would compare to ionic radius of Li^+ ? What about F to F^- ? How do you think the addition or loss of electrons influences the force of attraction between valence electrons and protons in the nucleus? What about the force of repulsion between valence electrons and the inner core electrons?

Table 3. Electron configuration and Bohr model for lithium and flu

S p e c i e s	Li	Li^+	F
E l e c t r o n c o n f i g u r a t i o n	$1s^2 2s^1$	$1s^2$	$1s^2 2s^2 2p^5$

B o h r m o d e l	 More information	 More information	 More information
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Positive ions

- Smaller in size than their parent atom.
- Loss of electrons (typically all electrons) from outermost energy level.
- Increased attraction between nucleus and valence shell electrons.
- The greater the number of electrons removed, the smaller the ionic radius.

Negative ions

- Larger in size than their parent atom.
- Gain of electrons into outermost energy level increases repulsion between electrons.
- Decreased attraction between nucleus and valence shell electrons.
- The greater the number of electrons gained, the larger the ionic radius.

You can find data for the ionic radii of common ions for each element in [section 10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-and-ionic-radii-of-the-elements-id-45112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-and-ionic-radii-of-the-elements-id-45112/>) of the DP chemistry data booklet.

Isoelectronic species

Can you think of anything that the species Na^+ , Mg^{2+} , Al^{3+} , N^{3-} , O^{2-} , F^- and Ne all have in common? They all have ten electrons and have the same electron configuration. Species that have the same electron configuration are said to be isoelectronic. How can we compare the sizes of species that have the same electron configuration? We can consider the effective nuclear charge. **Table 4** shows each of these species along with the number of protons and electrons they each have. How do you think the sizes of each of these species would compare?

Table 4. Comparison of protons and electrons for isoelectronic species.

Species	N^{3-}	O^{2-}	F^-	Ne	Na^+	Mg^{2+}	
Protons	7	8	9	10	11	12	
Electrons	10	10	10	10	10	10	
Atomic/ ionic radius	Decreasing radius						
							👁 More information

Since each of the species in **Table 4** has the same number of electrons, the species with the greatest number of protons will have the greatest effective nuclear charge. The greater the effective nuclear charge, the stronger the force of attraction between the nucleus and the outer electrons, therefore the smaller the radius. The lower the effective nuclear charge, the weaker the force of attraction between the nucleus and the outer electrons, therefore the larger the radius. Therefore the order of decreasing radius for the species in **Table 4** would be $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Ne} > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$.

Use **Interactive 2** to order isoelectronic species from smallest to largest radius.

Interactive 2. Drag and Drop to Order Isoelectronic Species from Smallest to Largest Radius.

 More information for interactive 2

This interactive activity allows users to investigate isoelectronic species, which are atoms or ions that have the same number of electrons. The objective is to arrange these species by dragging and dropping them in the correct sequence from the smallest to the largest atomic radius.

The activity features two tables. The first table consists of three columns and four rows, with headers labeled smallest radius, in between, and largest radius. The four rows are left blank for users to fill in the appropriate species in the correct order.

The isoelectronic species in the second table are given below. This table has three rows and four columns. Each group is displayed in distinct colored sections:

Row 1. H^- , He, Li^+ (Purple Section)

Row 2. Na^+ , Mg^{2+} , Al^{3+} (Red Section)

Row 3. P^{3-} , S^{2-} , Cl^- (Orange Section)

Row 4. Ca^{2+} , Ar, P^{3-} (Yellow Section)

Read below for the solution:

The isoelectronic species for the first table in the order of smallest radius, in between, and largest radius is given below for the four rows.

Row 1: Li^+ , He, H^-

Row 2: Al^{3+} , Mg^{2+} , Na^+

Row 3: Cl^- , S^{2-} , P^{3-}

Row 4: Ca^{2+} , Ar, P^{3-}

Through this interactive, users will gain insights into how atomic radius varies among isoelectronic species and will practice correctly positioning them according to their sizes.

Activity

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:** Communication skills — Applying interpretive techniques to different forms of media
- **Time required to complete activity:** 45 minutes
- **Activity type:** Groups of 4—5 students

Create a display showing the atomic and ionic radii to scale for elements $Z = 1$ to $Z = 36$. The sizing and choice of materials is up to you. You could choose to use paper, modelling clay, balloons, pie plates, balls, yarn, etc as materials to create your display. Your scale can be either 2-D or 3-D. Be sure to clearly indicate the parent atom and the ion for each element.

After each group has created their display, perform a peer evaluation of each group's display. The group with the highest scoring peer evaluation can have their radii display permanently displayed in the classroom. Use the following rubric to evaluate the display.

	3 points	2 points
Creativity	Unique with design elements that effectively convey the radii of elements and ions.	Design elements that effectively convey the radii of elements and ions.
Visual appeal	Visually appealing, well-organised and easy to understand.	Somewhat visually appealing and/or somewhat easy to understand.
Accuracy	Sizes of atomic and ionic radii are accurate to scale.	Sizes of atomic and ionic radii are mostly accurate to scale.

Periodicity of ionisation energy, electron affinity and electronegativity

Learning outcomes

By the end of this section you should be able to:

- State the definition for the terms ionisation energy, electron affinity and electronegativity.
- Explain the periodicity of ionisation energy, electron affinity and electronegativity.

When we studied ionic bonding in [section S2.1.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) we looked at how metals have a tendency to lose electrons to form positive ions and non-metals have a tendency to gain electrons to form negative ions. But how might we quantify these tendencies? How might we compare the tendencies to lose or gain electrons between atoms?

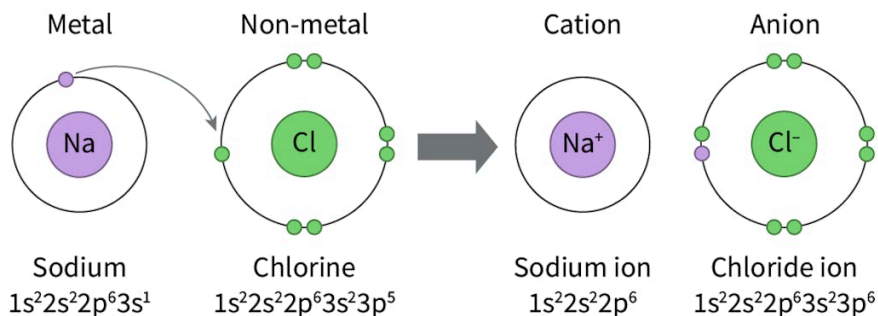


Figure 1. Metals have tendencies to lose electrons and non-metals have tendencies to gain electrons.

📄 More information for figure 1

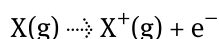
In **Figure 1** you can see the formation of the sodium cation. You will notice that the valence electron from the $n = 3$ energy level has been removed to form this cation. How is this electron removed? What is needed for the removal of an electron from an atom?

You may recall from [section S1.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) that when an electron absorbs enough energy to be promoted to the $n = \infty$ energy level, this

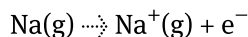
represents the point when the electron has been removed from the atom. This is when enough energy has been applied to overcome the attraction between the nucleus and that electron. The atom is said to be ionised once the removal of an electron occurs.

Ionisation energy

Ionisation energy is the energy required to remove the outermost electron from a species. A more specific form of this term is the first ionisation energy. This is the minimum energy required to lose one mole of electrons from one mole of gaseous atoms in the ground state. The qualifiers 'one mole' and 'gaseous' in this definition are used to keep variables constant to directly compare between different elements. The general equation for the first ionisation energy of an element is as follows:



Using sodium, Na, as an example, the equation is:



Data for the first ionisation energy for all elements can be found in [section 9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet. The first ionisation energies are plotted for the first 20 elements in **Figure 2**. Using this graph, identify the elements from the same group (e.g. group 1 H (Z = 1), Li (Z = 3), Na (Z = 11), K (Z = 19)). How would you describe the trend in first ionisation energy going down a group? How would you describe the trend in first ionisation energy radius going across a period? Can you think of explanations for these trends?

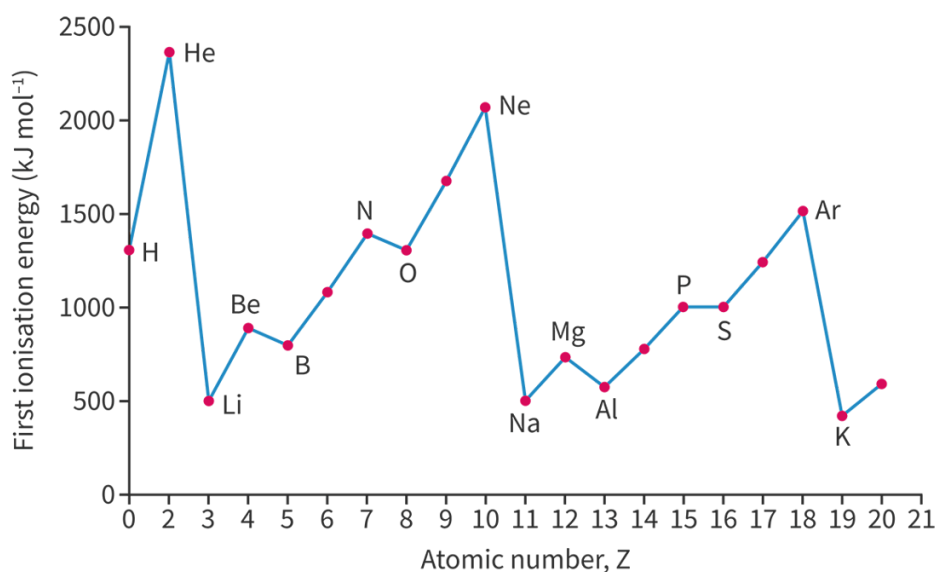


Figure 2. First ionisation energy for elements for the first 20 elements.

More information for figure 2

Table 1 shows the first ionisation energies for elements in groups 1, 2, 17 and 18. Looking at the atomic radii illustrated in **Figure 3**, how does the trend of first ionisation energies compare with that of atomic radii? What do you notice about the trend in first ionisation energy going down a group?

Table 1. First ionisation energy for elements in groups 1, 2,

Period	Highest occupied energy level	Group 1 First ionisation energy (kJ mol^{-1})	Group 2 First ionisation energy (kJ mol^{-1})	Group 17 First ionisation energy (kJ mol^{-1})	Group 18 First ionisation energy (kJ mol^{-1})
1	$n = 1$	H 1312			
2	$n = 2$	Li 520	Be 900	F 1681	Ne 2081
3	$n = 3$	Na 496	Mg 738	Cl 1251	Ar 1782
4	$n = 4$	K 419	Ca 590	Br 1139	Kr 1951
5	$n = 5$	Rb 403	Sr 549	I 1012	Xe 1980
6	$n = 6$	Cs 376	Ba 503	At 840	Rn 1936

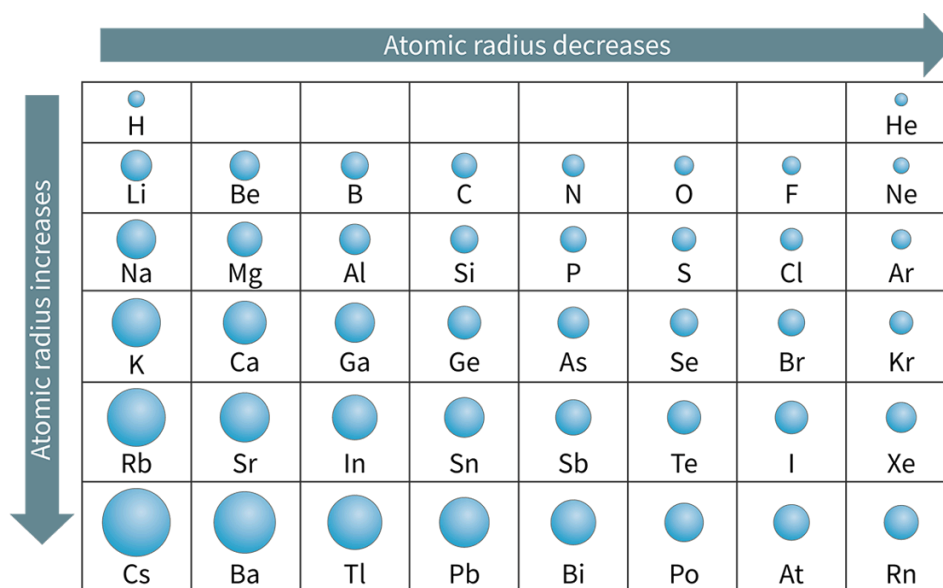


Figure 3. Atomic radii of the elements.

More information for figure 3

Table 2 shows the number of protons, atomic radius and first ionisation energy for period 3 elements. Why would the first ionisation energy increase across the period when valence electrons are all occupying the same $n = 3$ energy level?

Table 2. Number of protons, atomic radius and first ionisation energy for period 3 elements

Element	Na	Mg	Al	Si	P
# of protons	11	12	13	14	15
Atomic radius ($\times 10^{-12}$ m)	160	140	124	114	109
First ionisation energy (kJ mol^{-1})	496	738	578	786	1012

General trend in first ionisation down a group

By analysing **Table 1** and **Figure 3**, you can compile the following general trends in first ionisation energy going down a group:

- Going down a group the first ionisation energy decreases.
- Going down a group atomic radii increases and electrons occupy higher energy levels.
- Electrons in higher energy levels have a weaker attraction to the nucleus.
- Less energy is required to remove an electron.

General trend in first ionisation energy across a period

By analysing **Table 2**, you can compile the following general trends in first ionisation energy going across a period:

- Going across a period the first ionisation energy increases.
- Going across a period the nuclear charge increases (there are more protons in the nucleus).
- Going across a period the atomic radius decreases (atoms get smaller except for Ar which increases slightly compared to Cl).

- More energy is required to remove an electron.

So, to summarise, the trend in ionisation energy across a period depends on the nuclear charge and the atomic radius. The greater the nuclear charge and the smaller the atomic radius, the stronger the attraction between the nucleus and the valence electrons and the higher the ionisation energy. The trend in ionisation energy going down a group depends on the atomic radius and the number of occupied energy levels. The more occupied energy levels, the greater the atomic radius which results in a weaker attraction between the nucleus and the valence electrons. This causes the ionisation energy to decrease down a group.

Making connections

You may have noticed here that we are only evaluating general trends (i.e. the general increase or decrease across a period). There are discontinuities within these trends that we will look at more closely in [section S1.3.6](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) and [section S3.1.7](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/discontinuities-in-ionisation-energy-hl-id-44697/>).

Test your knowledge of the trends in ionisation energy by placing a > or < sign in the boxes below to indicate which element has the greater first ionisation energy. For example, $\text{Mg} > \text{Ca}$ would be used to indicate that Mg has a greater first ionisation energy than Ca.

Interactive 1. Identification of which element has the greater first ionisation energy.

More information for interactive 1

Electron affinity

The term ionisation energy is specific to the absorption of energy to remove an electron to form positive cations. The quantitative term that is specific to the gain of electrons to form negative ions is called electron affinity. If you think about the term affinity, what do you think it might mean? A non-scientific understanding of the word affinity would be to have a natural liking for. You can think of electron affinity in a similar manner, elements with a large absolute value for electron affinity are those with the greatest tendency to gain electrons and those with the lowest absolute value for electron affinity are those with the lowest tendency to gain electrons.

Electron affinity is the energy change that occurs when an electron is added to a species. When a neutral atom gains an electron, energy is released. This is called the first electron affinity and is illustrated in **Figure 4** for hydrogen. Energy is released due to the small force of attraction between the nucleus and the incoming electron.

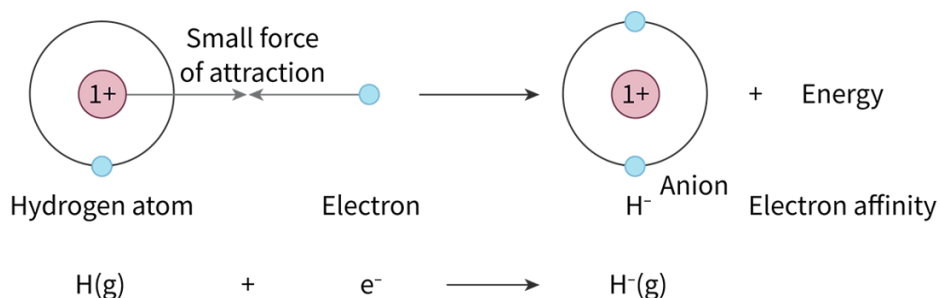


Figure 4. Hydrogen releasing energy as it gains an electron.

More information for figure 4

Section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet has values for the first electron affinity of most elements. You will notice that these values are negative, indicating the release of energy. **Table 3** shows the first electron affinity for group 17. How would you describe the general trend of electron affinity down a group?

Table 3. First electron affinity values for group 17 elements.

Group 17	
Element	Electron affinity (kJ mol^{-1})
F	—328
Cl	—349
Br	—325
I	—295
At	—270

General trend in electron affinity going down a group

- The absolute value for first electron affinity generally decreases going down a group (lower affinity for gaining an electron).
- Elements further down have a larger atomic radius, with outer valence shells further away from the nucleus.
- Weaker attraction between the added electron and the nucleus.

Table 4 shows the first electron affinity for period 3 elements. How would you describe the general trend of electron affinity across a period? Why do you think Mg and Ar have no values for electron affinity?

Table 4. Electron affinity values for period 3 elements.

Element	Na	Mg	Al	Si	P	S	Cl
Electron affinity (kJ mol^{-1})	—53		—42	—134	—72	—200	—329

General trend in electron affinity going across a period

- Absolute value for first electron affinity generally increases going across a period (higher affinity for gaining an electron).

- Elements across the period have a higher effective nuclear charge.
- Stronger attraction between the added electron and the nucleus.

You may have noticed that there are no values for first electron affinity listed for Mg and Ar. This is because many elements with completely filled subshells do not tend to gain electrons, therefore they do not have an associated electron affinity value.

Test your knowledge of the trends in electron affinity by placing a > or < sign in the boxes below to indicate which element has the greater first electron affinity.

Identify which element has the greater first electron affinity.

Use < or > to fill in the blanks.

First electron affinity

S Na

O F

Li F

Na K

Cl Br

 Check

Interactive 2. Identification of which element has the greater first electron affinity.

 More information for interactive 2

Electronegativity

Recall from section S2.2.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>) that electronegativity is the tendency of an atom to attract a shared pair of electrons in a covalent bond. Each atom has an assigned electronegativity value that is relative to fluorine. They can be found in section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet. Of the quantities discussed in this section, which quantity

would you expect the trends in electronegativity to be most similar to? What do you think the general trend for electronegativity would be going down a group? What about going across a period?

Table 5 shows the electronegativity values for group 17. How would you describe the general trend of electronegativity going down a group?

Table 5. Electronegativity values for group 17 elements.

Group 17	
Element	Electronegativity
F	4.0
Cl	3.2
Br	3.0
I	2.7
At	2.2

General trend in electronegativity going down a group

- Electronegativity values generally decrease down a group
- Atomic radius increases down a group from additional energy levels
- Weaker attraction between the nucleus and the shared pair of electrons

Table 6 shows the electronegativity values for period 2 elements. How would you describe the general trend across a period? Why do you think the noble gases do not have an electronegativity value assigned to them?

Table 6. Electronegativity values for period 2 elements

Element	Li	Be	B	C	N
Electronegativity	1.0	1.6	2.0	2.6	3.0

General trend in electronegativity going across a period

- Electronegativity values generally increase going across a period.
- Effective nuclear charge increases going across a period.
- Stronger attraction between the nucleus and the shared pair of electrons.

Aspect: Evidence

Electronegativity values are relative and are a 'made up' quantity. These values are calculated by considering the energy it takes to break the bonds between two elements. We assign a lot of value to these numbers (think back to the bonding triangle) but the fact that they are not measurable quantities certainly has its limitations. When scientists encounter evidence that does not fit a model, it calls for both reexamination of the evidence and reexamination of the model. In April 2021, scientists proposed an updated scale for electronegativities in [this](https://www.nature.com/articles/s41467-021-22429-0)  (<https://www.nature.com/articles/s41467-021-22429-0>) *Nature Communications* paper.

The noble gases do not have an assigned value for electronegativity as they do not form shared pairs of electrons with other atoms.

Test your knowledge of the trends in electronegativity by placing a > or < sign in the boxes below to indicate which element has the greater electronegativity.

Interactive 3. Identification of which element has the greater electronegativity.

👁 More information for interactive 3

Periodicity of atomic radius, ionisation energy, electron affinity and electronegativity

In both this and the previous [section 3.1.3a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>), we have been looking at general trends in the periodic table for atomic radius, ionisation energy, electron affinity and electronegativity. **Figure 5** is a periodic table summarising these general trends.

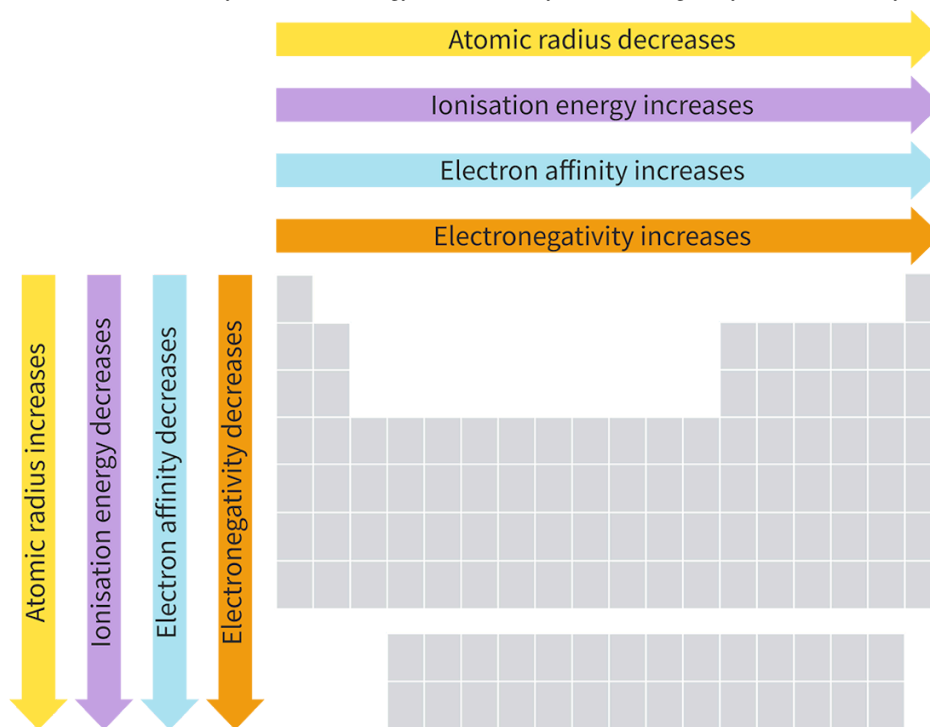


Figure 5. General trends for atomic radius, ionisation energy, electron affinity and electronegativity in the periodic table.

📄 More information for figure 5

Study skills

Periodicity refers to the general trends we observe in the periodic table for different properties. While it is helpful to describe the trends in terms of increasing/decreasing from left to right and top to bottom it is important to note that these are not explanations. Explanations for these trends involve unpacking the attractive forces between the nucleus and the outer valence shell, along with the repulsion between electrons. The strength of the attractive forces is always influenced by the number of occupied energy levels and the effective nuclear charge of an atom.

Activity

- **IB learner profile attribute:** Balanced
- **Approaches to learning:** Research skills — Evaluating information sources for accuracy
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Group activity

Pasta periodic table

Materials needed (for each student):

- Dry spaghetti (alternatively can use bamboo skewers or pipe cleaners)
- Ruler
- Glue
- Printed copy of the periodic table
- Two pieces of cardboard the same size as your periodic table
- Thumb tack (to poke holes into paper and cardboard)

As a group of four, each group member will select one of the following periodic trends:

- Atomic radius
- Ionisation energy
- Electron affinity
- Electronegativity

You will create a three-dimensional bar graph for your trend using spaghetti to represent the bars. You will need to scale the length of each spaghetti stick to represent value of your chosen parameter. Below provides steps to complete this task.

Three-dimensional bar graph

- Glue both pieces of cardboard together to create one thick piece.
- Glue the periodic table on the top side.
- Using the thumb tack poke a hole through the periodic table and the first cardboard layer for each element.
- Measure the length of your spaghetti stick.
- Decide what measurement you would like each unit to represent. For example, if a spaghetti stick measured 25 cm, then your largest value can only be 25 cm. Let's say the largest value you are representing is 9567. This would be represented by the full spaghetti stick. Now if you wanted to represent 8653 using spaghetti sticks you would simply multiply 25 by $\frac{8653}{9567}$.
- Measure and cut your spaghetti to scale for each value using the following formula:

$$\text{length of full spaghetti} \times \frac{\text{value you are representing}}{\text{largest value from your data set}}$$

- Insert each stick of spaghetti into the appropriate element punched hole in your periodic table.

Once you are finished with your periodic table, help your group members with their spaghetti sticks until all group members are finished. Place all periodic tables side by side.

As a group discuss and complete the following:

- Identify any periodic trends that are evident.
- Discuss the ease to which you were able to identify trends. Were any properties particularly difficult to identify a clear trend? Share why you think these are difficult.
- In ChatGPT ask the AI chatbot to explain each trend down a group and across a period. The AI chatbot will likely produce a reasonable but incomplete response. As a group evaluate the response. What parts of the

response were well crafted? What parts of the response are missing for a high-level chemistry explanation?